(Please write your Enrollment No. immediately)

Enrollment No:.....

SUPPLEMENTARY MID-TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

Second Semester, March 2025

Paper Code: **BS106**

Subject: Applied Physics-II

Time: 1½Hrs.

Maximum Marks: 30

Note: Attempt Question no 1, which is compulsory, and any two more questions from the remaining

Q. No.	Question 1	Marks	CO
1(a)	Calculate the De-Broglie wavelength of 40KeV electron	[2.5]	CO1
1(b)	Find the average speed and root mean square speed for an assembly of two molecules, one with a speed of 1m/s and the other with a speed of 3m/s.	[2.5]	CO2
1(c)	Give physical interpretation of wave function ψ and its normalization.	[2.5]	CO1
1(d)	The Fermi level in potassium is 2.1 eV at a particular temperature. Calculate the number of free electrons per unit volume in potassium at the same temperature.	[2.5]	CO2
	Question 2		
2(a)	Set up the Schrodinger wave equation for a particle in an infinite well (1D). Obtain the wave functions and energy eigenvalues.	[5]	CO1
2(b)	What is Quantum Mechanical Tunneling	[2]	CO1
2(c)	An electron has a speed 500m/s accurate to 0.01%. With what fundamental accuracy can we locate the position of an electron?	[3]	CO1
	Question 3		
3(a)	What is Black Body Radiation? Derive Planck's formula for the energy distribution in the spectrum of a Black body.	[6]	CO2
3(b)	Fermi Energy in Ag is 5.51eV. What is the average energy of free electrons in Ag at 0K? What temperature is necessary for the average molecular energy in an ideal gas to have this value?	[4]	CO2
	Question 4		
4(a)	Compare the salient features of Maxwell Boltzmann(MB), Bose Einstein(BE) and Fermi Dirac(FD) Statistics	[5]	CO2
4(b)	Calculate the expectation value of the momentum of a particle trapped inside a 1D box.	[2]	CO1
4(c)	What is zero point energy in quantum mechanics? The energy of the linear Harmonic Oscillator in the third excited state is 0.1eV. Find the frequency of vibration.	[3]	CO1

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END TERM EXAMINATION

FIRST SEMESTER [B.TECH] JANUARY 2024

Paper Code: BS-103 **Subject: Applied Chemistry**

Time: 3 Hours **Maximum Marks: 60**

Note: Attempt five questions in all including Q. No.1 which is compulsory. Internal Choice is indicated. Assume missing data,if any.

Q1 Attempt all questions:- **(2.5x8=20)**

(a) Calculate the LCV of a fuel which has 9% hydrogen & HCV is 6250 cal/g. Given the latent heat of steam is 587cal/g.

(b) Write a short note on proximate analysis.

(c) Explain the term component. How many components are present in the following system?

(i) Water ⇌ water vapour

(ii) KCl + Water ⇌ KCl hydrate

(d) Give the mechanism of addition & condensation polymerization.

(e) Determine the $CaCO_3$ equivalent of 83 mg of Mg $(HCO_3)_2$. (Mol. Weight =146).

(f) What is Calgon conditioning? How scale formation can be prevented by using this internal water treatment method?

(g) Impure metal corrode faster than pure metal under identical conditions. Give reason.

(h) Define Zero Waste technology with a suitable example.

Q2 **Attempt any one part between I & II.**

I (a) Give an account of the determination of calorific value of gaseous fuel by Boy's gas calorimeter with the help of a neat diagram. **(4)**

(b) How is compressed natural gas (CNG) different from the liquified natural gas (LPG). **(3)**

(c) Out of straight-chain hydrocarbons & aromatics, which one would have high anti-knocking properties? **(3)**

OR

II (a) A sample of coal containing 92% C,5% H , 3% ash , when coal was tested in the laboratory for its calorific value in bomb calorimeter, the following data were obtained:- **(6)**

Weight of coal burnt =0.95 gm, Weight of water taken =700 gm,
Water equivalent of bomb & calorimeter =2000 gm
Rise in temperature =2.48° C, Cooling correction =0.2° C,
Fuse wire correction =10 cal, Acid correction =60.0 cal
Calculate the gross & net calorific values of the coal in cal/gm. (Latent heat of condensation of steam as 580 cal/gm).

(b) Describe the manufacture of metallurgical coke by Otto Hoffmann process. **(4)**

Q3 **Attempt any one part between I & II.**

I (a) State phase rule & define the terms involved. Explain Sulphur system based on phase rule. **(4)**

(b) Write the structural unit & two important applications of each of the following:- **(3)**

(i) Urea-formaldehyde

(ii) Nylon 6,6

(iii) PVC

(c) What is ultra-thin molecular weight polythene? Write their unique properties & applications. **(3)**

P.T.O.

OR

II (a) Define conducting polymer, their types & applications of conducting polymer in engineering. **(6)**

(b) Give an example of a two-component system which form compound with congruent melting point. **(4)**

Q4 **Attempt any one part between I & II.**

I (a) Calculate the amount of Lime (92% pure) & Soda (98% pure) required for the treatment of 50,000 liter of water whose analysis is as follows: **(4**
$Ca(HCO_3)$= 42.5ppm , **$Mg(HCO_3)$** = 36.5 ppm, **$MgSO_4$** = 30 ppm, **$CaSo_4$** = 34.0 ppm ,**$CaCl_2$** = 27.75 ppm, **NaCl**=10.0ppm

(b) Write the important steps involved in the defluoridation of water. **(3)**

(c) How the hardness of water is determined by EDTA titration method? **(3)**

OR

II (a) Explain with reactions & diagram, the ion-exchange resins process for the softening of water. How can they be regenerated after getting exhausted? **(6)**

(b) A sample of water was alkaline both phenolphthalein and methyl - orange alkalinity. 100 ml of this water sample required 20 mL of 50 H_2SO_4 for the phenolphthalein endpoint and 15 mL for complete neutralization. Calculate the type of alkalinity in ppm. **(4)**

Q5 **Attempt any one part between I & II**

I (a) Explain the mechanism of hydrogen evolution & oxygen absorption in electrochemical corrosion. **(4)**

(b) Discuss the Atom economy concept. Calculate atom economy in an addition reaction. **(3)**

(c) Discuss the sacrificial anodic protection. What is the condition for a metal to act as a sacrificial anode to iron? **(3)**

OR

II (a) Explain the underlying principles of the BET method and how it measures the surface area of nanoparticles. Explain the applications of Nanotechnology in different fields. **(6)**

(b) What are the twelve principles of green chemistry, and how do they guide sustainable chemical practices? **(4)**

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END TERM EXAMINATION

FIRST SEMESTER [B.TECH] JANUARY 2024

Paper Code: BS-105 ***Subject: Applied Physics-I***

Time: 3 Hours ***Maximum Marks: 60***

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit. Symbols have their usual meanings. Assume missing data, if any.

Q1 Attempt all questions:

a) Show that work done for a perfect gas can be given by $W = C_V(T_i - T_f)$ for adiabatic process. **[2]**

b) Distinguish between displacement and conventional current. **[2]**

c) Find the resultant of superposition of two waves $Y_1 = 2 \sin \omega t$ and $Y_2 = 5 \sin(\omega t + 30^0)$. Symbols have their usual meanings. **[2]**

d) At what speed will an object of length 100 cm be measured as 50 cm to an observer at rest. **[2]**

e) Explain the difference between spatial and temporal coherence. **[2]**

f) Why the grating of large number of lines are preferred? **[2]**

UNIT-1

Q2 a) Prove that the entropy of an ideal gas remains constant in a reversible process. But it is increases in an irreversible process. **[6]**

b) State first law of thermodynamics and show that heat and work are path functions but their difference is a point function. **[4]**

c) Discuss continuum model of thermodynamics. **[2]**

Q3 a) Prove that the efficiency of a Carnot's engine depends only upon the two temperatures between which it works. **[6]**

b) What are the limitations of first law of thermodynamics. State second law of thermodynamics. **[4]**

c) A Carnot's refrigerator absorbs heat from water at 0^0 C and rejects it at room temperature 37^0 C. Calculate the amount of work required to convert 10 kg water at 0^0 C into ice at same temperature [latent heat of ice = 3.4×10^5 J/kg]. Also find the coefficient of performance of the refrigerator. **[2]**

UNIT-II

Q4 a) Prove that the speed of plane electromagnetic wave in isotropic dielectric is less than the speed of electromagnetic wave in free space. Also prove the orthogonality of E, H and k. **[6]**

b) Set up continuity equation and discuss its physical significance. **[4]**

c) Calculate the magnitude of pointing vector at the surface of the sun. Given that power irradiated by the sun = 3.8×10^{26} watt and radius of sun = 7×10^8 m. **[2]**

Q5 a) Write Maxwell's equation in integral and differential forms. Discuss in brief the physical meaning of each of these. Also derive Maxwell's third equation in differential form. **[6]**

b) Derive velocity and energy in SHM. Also give graphical representation of it. **[4]**

c) Calculate the skin depth for 3 MHz electromagnetic wave through copper. [given conductivity $\sigma = 6 \times 10^7 mho/m$, $\mu = 4\pi \times 10^{-7} Henry/m$]. **[2]**

UNIT-III

Q6 a) Explain the formation of fringes in Newton's ring experiment. Give its application to find out wavelength of light. **[6]**

b) Explain the phenomena of double refraction. Describe the working principle of a Nicol prism. How is Nicol prism used to produce circularly polarised light. **[4]**

c) Draw a labelled ray diagram depicting interference by biprism. **[2]**

Q7 a) Derive an expression for intensity of diffracted light in Fraunhofer diffraction at a single slit. **[6]**

b) Explain with a series of neat well labelled diagrams the functioning of the retarding plates: (i) Half wave plate (ii) Quarter wave plate. **[4]**

c) Each slit has a width of 0.15 mm and distance between their centre is 0.75 mm. What are the missing order. **[2]**

UNIT-IV

Q8 a) Describe Michelson Morley experiment. Explain its main conclusions. **[6]**

b) Using Lorentz transformation equations, prove that "Moving clocks appear to go slow". **[4]**

c) Deduce the expression $E=mc^2$. **[2]**

Q9 a) Discuss Einstein's transition probabilities and derive the relation between Einstein's A and B coefficients. **[6]**

b) Describe the working of He-Ne LASER, explain its energy level diagram. **[4]**

c) What is optical pumping? How does it help in achieving population inversion in a LASER. **[2]**

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